

Experiment No-01

Aim: Write an Assembly language program to find sum of 10 number present in memory and store the result into memory.

Software Used: Keil uVision5

Program:

```
                AREA addition, CODE, READONLY
                EXPORT __main
__main
                MOVS R0, #09
                LDR R1, =0X1FFFF000
                LDRB R2, [R1]
UP
                ADDS R1, #01
                LDRB R3, [R1]
                ADDS R2, R3
                SUBS R0, #01
                BNE UP

                ADDS R0, #01
                STRB R2, [R1]
                END
```

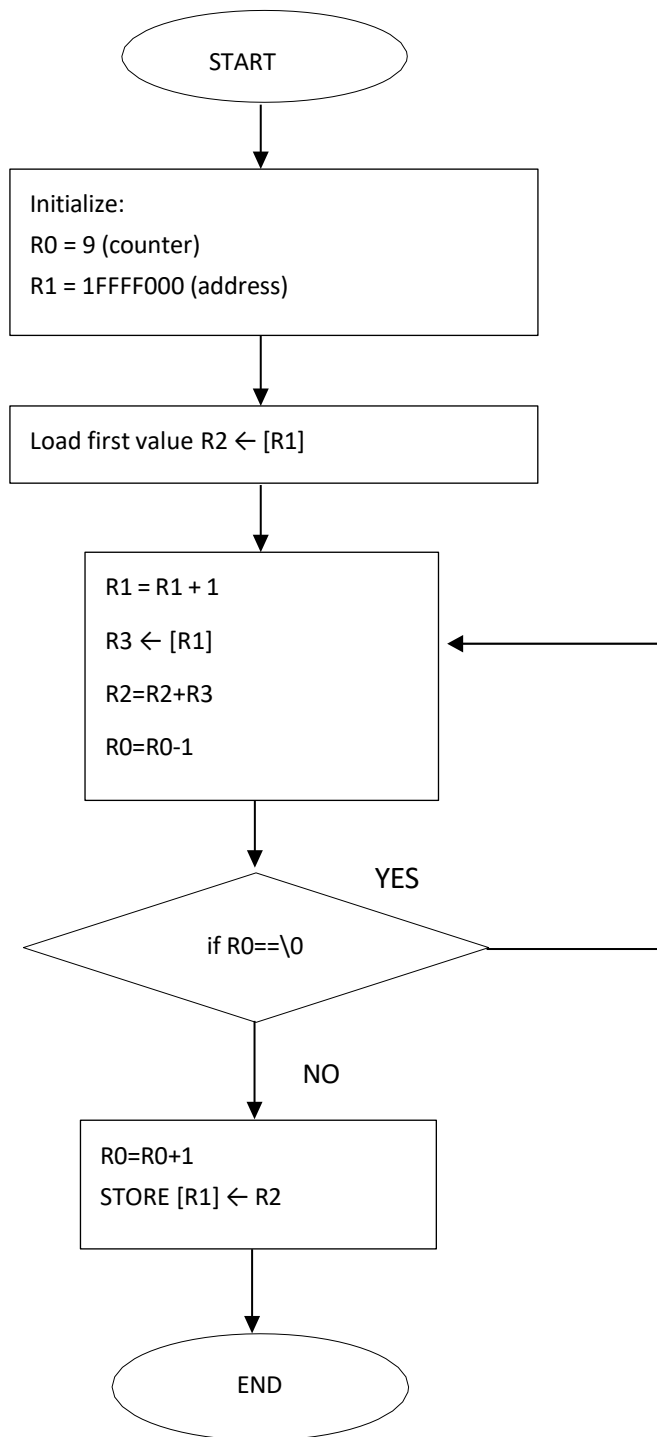
Program Screenshot:

The screenshot displays the uVision IDE interface with the following components:

- Registers:** A list of registers (R0-R15, xPSR) and their values. R15 (PC) is highlighted with a value of 0x0000124.
- Disassembly:** A window showing assembly code. The instruction `ADDS r2,r2,r3` is highlighted in yellow. The code includes a `break;` statement and a `case 0x80U:` block.
- Command:** A window showing the command line and error messages. The error message is: `*** error 65: access violation at 0x40048100 : no 'write' permission`.
- Memory:** A window showing memory addresses and their values. The address 0x1FFFF000 is highlighted.

The status bar at the bottom indicates the simulation is running at `t1: 0.00000717 sec` with `L7 C:1` and `CAP NUM SCRL OVR R/W`.

Flowchart:



Results and Conclusion:

Output:

Registers	
Register	Value
Core	
R0	0x00000000
R1	0x1FFFF009
R2	0x00000127
R3	0x00000040
R4	0x00000000
R5	0x00000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x00000000
R11	0x00000000
R12	0x00000000
R13 (SP)	0x1FFFF100
R14 (LR)	0x000000C7
R15 (PC)	0x00000124
+ xPSR	0x61000000
+ Banked	
+ System	
- Internal	
Mode	Thread
Privilege	Privileged
Stack	MSP
States	86
Sec	0.00000717

Memory:

Memory 2	
Address:	0x1FFFF000
0x1FFFF000:	12 13 01 06 19 78 09 11 10 40 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0x1FFFF01C:	00 00
0x1FFFF038:	00 00
0x1FFFF054:	00 00
0x1FFFF070:	00 00
0x1FFFF08C:	00 00
Call Stack + Locals Memory 1 Memory 2	
Simulation	
t1: 0.00000717 sec	
L:7 C:1	
CAP NUM SCRL OVR R/W	

10 Numbers are added and the sum is available in Register R1 and Memory